

STOKKA, Norway

WMO: 011160

Lat: 65.961N

Lon: 12.471E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.2	19.4	-0.3	5.4	21.4	4.0	6.8	23.5	34.5	39.1	31.6	39.4	9.0	70	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	75.8	62.8	71.6	60.4	68.0	58.2	64.5	73.9	61.9	69.5	59.6	65.9	7.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	79.4	68.3	58.6	73.5	65.4	56.8	68.9	62.5	29.5	74.3	27.7	69.8	26.1	66.1	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 29.4	25.3	21.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	10.6	67.0	4.4	3.5	7.5	69.6	4.9	71.6	2.5	73.6	-0.7	76.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	32.6	32.3	33.9	39.4	46.6	51.0	57.3	56.7	52.1	44.6	37.9	35.7
	DBStd	10.72	6.89	6.52	5.75	5.66	6.61	5.46	6.41	5.22	5.34	5.69	6.37	5.95
	HDD50	3084	538	495	498	323	150	47	3	3	37	184	363	443
	HDD65	7909	1003	915	963	767	572	424	258	266	388	633	812	908
	CDD50	680	0	0	0	6	44	76	229	209	99	16	1	0
	CDD65	30	0	0	0	0	1	2	19	8	0	0	0	0
	CDH74	206	0	0	0	0	0	7	161	38	0	0	0	0
	CDH80	30	0	0	0	0	0	0	28	2	0	0	0	0
	WSAvg	9.8	11.1	11.0	11.3	9.8	9.1	9.1	7.9	8.1	8.4	10.0	10.7	11.8
	PrecAvg	54.3	5.2	4.7	4.5	3.6	3.0	3.3	3.5	3.7	6.1	6.1	5.2	6.0
(15) Precipitation	PrecMax	75.4	8.8	9.6	8.4	7.4	4.7	6.8	8.9	8.9	14.6	12.5	9.8	13.1
	PrecMin	41.3	0.0	0.7	1.3	0.9	0.8	0.6	1.2	0.6	1.6	1.4	0.4	0.0
	PrecStd	10.4	2.4	2.5	2.1	1.7	0.9	1.7	2.3	1.9	3.3	3.2	3.0	3.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	48.3	46.0	49.2	64.7	73.7	75.1	83.7	78.7	70.0	62.6	55.3	49.4	
	0.4% MCWB	39.2	41.0	40.2	49.5	58.0	59.4	66.8	64.5	58.9	53.4	48.8	41.9	
	2% DB	45.1	44.0	45.5	55.8	68.2	69.7	78.1	73.7	66.3	58.4	50.4	46.8	
	2% MCWB	42.0	40.6	40.5	45.5	55.3	57.5	65.2	61.9	56.2	50.4	45.4	43.2	
	5% DB	43.4	42.6	43.7	51.5	63.5	64.8	73.3	69.9	63.9	55.2	48.6	45.4	
	5% MCWB	40.6	39.9	40.2	43.3	53.0	55.1	63.3	60.3	55.1	48.8	44.5	42.3	
	10% DB	42.3	41.0	42.1	48.0	58.3	60.7	68.1	66.5	60.9	52.2	46.6	44.0	
	10% MCWB	39.5	38.2	38.9	41.5	50.2	53.5	59.8	58.3	54.0	47.9	43.0	40.7	
	0.4% WB	44.9	43.5	44.0	49.8	59.5	60.7	69.1	66.0	61.0	54.0	49.9	46.5	
	0.4% MCDB	45.6	44.0	44.9	61.5	71.0	71.2	80.3	75.9	66.0	59.5	51.3	46.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	43.1	42.1	42.7	47.1	55.8	58.1	65.8	63.3	58.2	51.9	47.2	44.9	
	2% MCDB	43.9	43.0	44.0	54.3	66.1	67.4	75.9	71.6	63.9	55.7	49.7	45.6	
	5% WB	41.4	40.4	41.0	44.4	53.5	56.1	63.6	61.0	56.4	50.2	45.3	43.2	
	5% MCDB	42.7	41.8	43.1	49.3	62.6	63.8	72.4	68.1	61.4	53.4	47.8	44.5	
	10% WB	39.3	38.3	39.0	42.5	50.9	53.8	60.7	59.1	54.9	48.7	43.2	40.9	
	10% MCDB	41.2	40.2	41.8	46.5	58.0	59.7	67.1	65.2	59.3	51.6	46.1	43.3	
(35) Mean Daily Temperature Range	MDBR	7.3	7.2	7.9	9.4	10.7	9.7	10.7	10.7	10.0	8.0	7.2	7.3	
	5% DB MCDBR	7.7	7.7	8.4	15.2	17.6	17.1	19.5	17.7	15.2	11.2	8.7	8.6	
	5% DB MCWBR	6.3	6.9	6.8	8.8	9.7	9.1	9.8	9.4	8.3	7.3	6.5	7.2	
	5% WB MCDBR	7.4	7.3	7.5	12.4	15.9	14.9	18.4	16.1	13.3	9.2	8.2	8.4	
	5% WB MCWBR	7.0	7.3	7.0	7.5	9.5	8.6	9.9	9.3	7.9	7.4	7.4	8.1	
(40) Clear-Sky Solar Irradiance	taub	0.368	0.303	0.330	0.354	0.354	0.350	0.358	0.345	0.333	0.309	0.347	0.364	
	taud	2.526	2.186	2.273	2.343	2.432	2.488	2.490	2.517	2.520	2.337	2.489	2.477	
	Ebn at Noon	68	174	231	256	270	273	267	260	235	174	64	14	
	Edh at Noon	7	17	25	30	30	30	29	25	20	15	6	3	
(44) All-Sky Solar Radiation	RadAvg	23	139	532	1053	1423	1552	1423	1097	586	229	44	4	
	RadStd	3	27	82	112	101	196	163	119	80	43	6	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-167	-239	N/A	N/A

Nomenclature: See separate page